

Challenge (Habanero)

Q1. Solve for x in the equation $2x + 5 = 11$.

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. Isolate y in the equation $y - 3 = 2x + 1$.

- (A) $y = 2x + 4$
- (B) $y = 2x - 4$
- (C) $y = x + 4$
- (D) $y = x - 4$
- (E) $y = 3x + 1$

Q3. A gamer's score is calculated by the formula $S = 2x + 10y$, where x is the number of levels completed and y is the number of coins collected. Solve for y .

- (A) $y = \frac{S-2x}{10}$
- (B) $y = \frac{S+2x}{10}$
- (C) $y = \frac{S-10x}{2}$
- (D) $y = \frac{S+10x}{2}$
- (E) $y = S - 2x$

Q4. A coding problem requires the equation $5x - 2 = 3x + 1$ to be solved for x .

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q5. In a tech competition, a team's score is given by $T =$

$4y - 3x$. Solve for x .

- (A) $x = \frac{4y-T}{3}$
- (B) $x = \frac{4y+T}{3}$
- (C) $x = \frac{3y-T}{4}$
- (D) $x = \frac{3y+T}{4}$
- (E) $x = 3y - T$

Q6. The cost of data usage is calculated by the formula $C = 0.5x + 2y$, where x is the number of gigs used and y is the number of months. Solve for x .

- (A) $x = \frac{C-2y}{0.5}$
- (B) $x = \frac{C+2y}{0.5}$
- (C) $x = 2C - 4y$
- (D) $x = 2C + 4y$
- (E) $x = C - 2y$

Q7. A scoreboard ranks players based on the formula $R = 3x - 2y$, where x is the number of wins and y is the number of losses. Solve for y .

- (A) $y = \frac{3x-R}{2}$
- (B) $y = \frac{3x+R}{2}$
- (C) $y = \frac{2x-R}{3}$
- (D) $y = \frac{2x+R}{3}$
- (E) $y = 2x - R$

Q8. The logic for a coding puzzle is given by $2x + 5 = 3y - 1$. Solve for x .

- (A) $x = \frac{3y-6}{2}$
- (B) $x = \frac{3y+6}{2}$
- (C) $x = \frac{2y-6}{3}$
- (D) $x = \frac{2y+6}{3}$
- (E) $x = 3y - 6$

Open Ended Questions (FRQ)

Q9. The formula to calculate the average data usage per month is $A = \frac{x}{y}$, where x is the total data used in gigs and y is the number of months. Solve for x and show your work. Assume $A = 2$ and $y = 4$.

Q10. The score of a game is determined by $S = x + 2y - 3z$, where x is the number of points earned, y is the number of levels completed, and z is the number of penalties. Solve for z and then find the value of z when $S = 10$, $x = 4$, and $y = 2$. Show your work.

Chef's Tips

- Emphasize the importance of isolating variables in algebraic expressions.
- Encourage students to check their work by substituting their answers back into the original equation.
- Remind students that literal equations can be solved using the same techniques as linear equations.